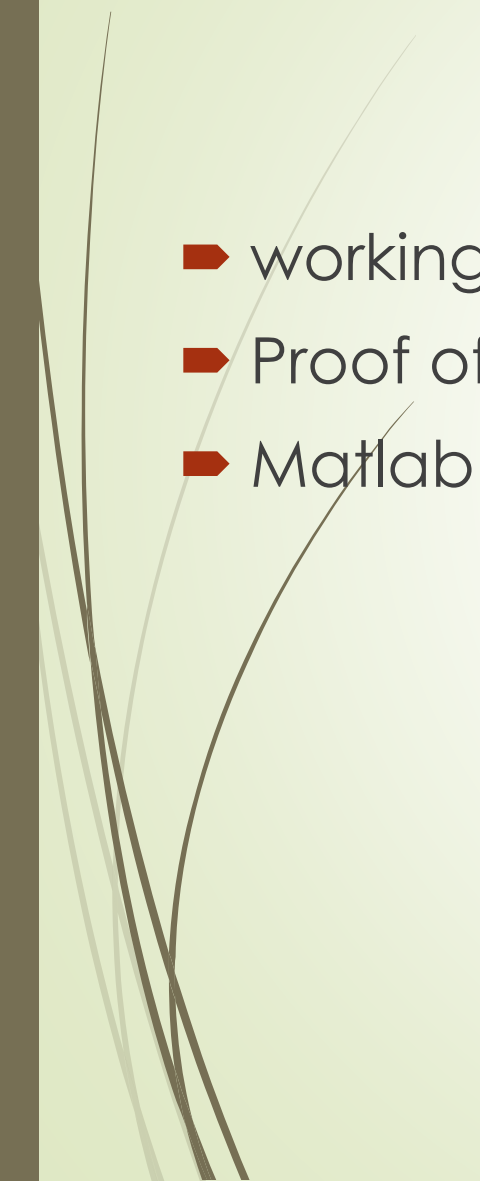




Microring modulators.



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- working principle
 - Proof of Transfer function
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- 

Microring modulators

working principle

- Resonance occurs when light circulating inside a ring-shaped waveguide exhibits constructive interferences.
- This happens only under a specific condition :The optical path length of the ring must be an exact multiple of the light's wavelength^[2].
- When the waves in the loop build up a round trip phase shift that equals an integer times 2π , **$\Phi = 2\pi m$** , the waves interfere constructively and the cavity is in resonance.

$$\phi = \frac{2\pi n_{\text{eff}} L}{\lambda}$$

$$\lambda_{\text{res}} = \frac{n_{\text{eff}} L}{m}$$

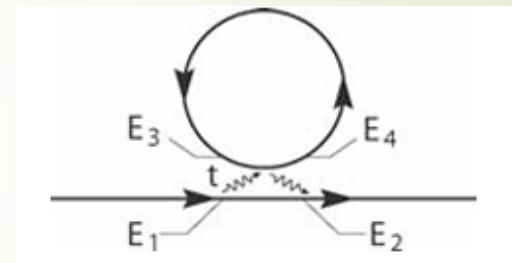
Microring modulators

Transfer function

-the fields output from the through port are related to the fields input to the waveguide by a scattering matrix:

$$\mathbf{b} = \mathbf{S} \mathbf{a}$$

$$\begin{pmatrix} E_4(\omega) \\ E_2(\omega) \end{pmatrix} = \begin{pmatrix} r & it \\ it & r \end{pmatrix} \begin{pmatrix} E_3(\omega) \\ E_1(\omega) \end{pmatrix}$$



-where the self- and cross-coupling coefficients r and t satisfy the relation $r^2 + t^2 = 1$.

-the incident field E_1 , transmitted E_2 , and circulating E_3 , E_4 fields where E_3 is the field in the ring **just before** coupling back into the bus (i.e., after one roundtrip and arriving at the coupler) and E_4 denotes The field in the ring just after the coupler (i.e. the field that will circulate and pick up the round-trip factor).

Microring modulators

Transfer function

-the lossless coupling equation ensures power conservation as no absorbed or radiated or scattered power :

-From
$$\begin{pmatrix} E_4(\omega) \\ E_2(\omega) \end{pmatrix} = \begin{pmatrix} r & it \\ it & r \end{pmatrix} \begin{pmatrix} E_3(\omega) \\ E_1(\omega) \end{pmatrix}$$

$$E_2 = it^*E_3 + r^*E_1 \text{-----}(1)$$

$$E_4 = r^*E_3 + it^*E_1$$

-But the field after one roundtrip E_3 is $ae^{-i\phi}$ times the field that was in the ring just after coupling E_4 .

$$\begin{aligned} E_3 &= ae^{-i\phi} E_4 \\ &= ae^{-i\phi} (r^*E_3 + it^*E_1) \end{aligned}$$

$$\frac{E_3}{E_1} = \frac{it^* ae^{-i\phi}}{1 - r^*ae^{-i\phi}} \text{-----}(2)$$

Microring modulators

Transfer function

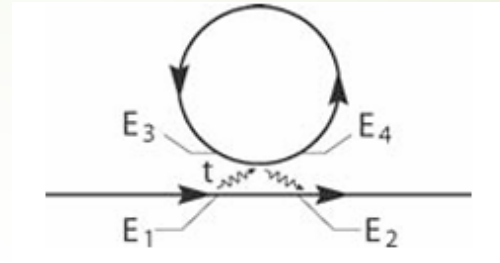
-from 1, 2

$$E_2 = r * E_1 + i t * \left(\frac{a e^{-i\phi}}{1 - r * a e^{-i\phi}} \right) E_1$$

$$\frac{E_2}{E_1} = \frac{-t^2 * a e^{-i\phi} - r^2 * a e^{-i\phi} + r}{1 - r * a e^{-i\phi}}$$

$$= - \frac{a e^{-i\phi} + r}{1 - r * a e^{-i\phi}}$$

$$= e^{i(\pi + \varphi)} \frac{a - r e^{i\phi}}{1 - r * a e^{-i\phi}}$$



Microring modulators

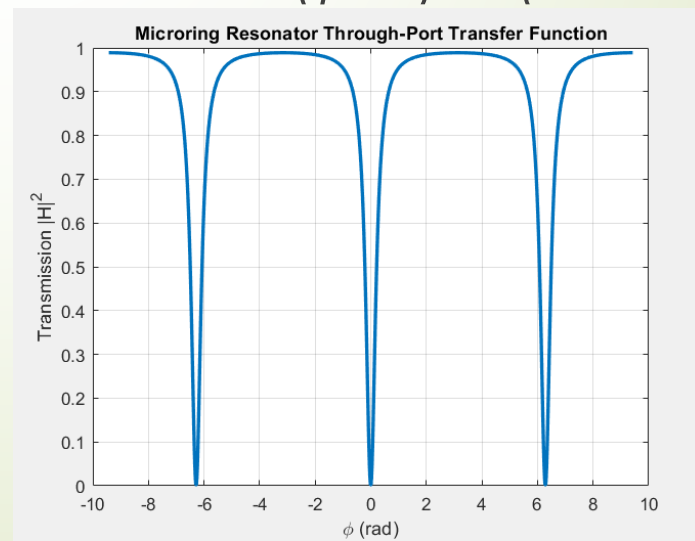
Transfer function

-the measurable power transmission intensity (through port)

$$T(\omega) = |H(\omega)|^2 = \frac{r^2 + a^2 - 2ar \cos \phi}{1 + a^2 r^2 - 2ar \cos \phi}$$

For $\phi=2\pi m$ (so $\cos \phi=1$), the numerator becomes $(r-a)^2$ and denominator $(1-ra)^2$.

Critical coupling occurs when $r=a$. Then $T(\phi_{res})=0$ (a deep notch at resonance).



References

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